Per-site ΔRMSE from adding mortality — C+T vs C+T+M

Below 0 = mortality reduces the site's RMSE | bars coloured by total mortality class (site maximum) | repeated CV (3×80 %) | tree cover ≥30 % (101 sites, 361 site-years) · deadtrees v5

XGBoost (Optuna) · 12m

XGBoost (Optuna) · 24m

LightGBM (Optuna) · 12m

LightGBM (Optuna) · 24m

GPPsat (µmol m⁻² s⁻¹)

57% improved

58% improved

56% improved

55% improved

2

0

-2

NEPmax (µmol m⁻² s⁻¹)

68% improved

70% improved

74% improved

68% improved

2

0

-2

ΔRMSE = RMSE(with M) − RMSE(without M)

ETmax (mm d⁻¹)

58% improved

64% improved

58% improved

57% improved

0.01

0.00

-0.01

-0.02

WUE (g C mm⁻¹)

53% improved

58% improved

53% improved

51% improved

0.8

0.4

0.0

-0.4

Sites (sorted by ΔRMSE within each panel)

Total mortality
site maximum, mean of 100-500 m buffers
(breaks: mean ± 0.5 SD)

- Low (< 10.6 %, 41 sites)
- Mid (10.6–24.0 %, 38 sites)
- High (> 24.0 %, 22 sites)